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RESEARCH ARTICLE

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Prediction of Alzheimer's disease progression within 6 years using speech: A novel approach leveraging language models

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Abstract

INTRODUCTION: Identification of individuals with mild cognitive impairment (MCI) who are at risk of developing Alzheimer's disease (AD) is crucial for early intervention and selection of clinical trials.

METHODS: We applied natural language processing techniques along with machine learning methods to develop a method for automated prediction of progression to AD within 6 years using speech. The study design was evaluated on the neuropsychological test interviews of $n = 166$ participants from the Framingham Heart Study, comprising 90 progressive MCI and 76 stable MCI cases.

RESULTS: Our best models, which used features generated from speech data, as well as age, sex, and education level, achieved an accuracy of 78.5% and a sensitivity of 81.1% to predict MCI-to-AD progression within 6 years.

DISCUSSION: The proposed method offers a fully automated procedure, providing an opportunity to develop an inexpensive, broadly accessible, and easy-to-administer screening tool for MCI-to-AD progression prediction, facilitating development of remote assessment.

KEYWORDS
Alzheimer's disease prognosis, Framingham Heart Study, natural language processing, neuropsychological test

Highlights

- Voice recordings from neuropsychological exams coupled with basic demographics can lead to strong predictive models of progression to dementia from mild cognitive impairment.
- The study leveraged AI methods for speech recognition and processed the resulting text using language models.
- The developed AI-powered pipeline can lead to fully automated assessment that could enable remote and cost-effective screening and prognosis for Alzehimer's disease.

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1 | BACKGROUND

Alzheimer's disease (AD) is the most common cause of dementia and has a long prodromal phase, during which subtle cognitive changes occur. Mild cognitive impairment (MCI) is a stage between normal cognition and AD. Individuals with MCI are at higher risk of developing AD with a 3% to 15% conversion rate of MCI to AD every year.[1,2] Therefore, accurately predicting the progression of MCI to AD can assist physicians in making decisions regarding patient treatment, participation in cognitive rehabilitation programs, and selection for clinical trials involving new drugs.[3]

Traditionally, AD pathology can be assessed using biomarkers such as cerebrospinal fluid assays or neuroimaging techniques like positron emission tomography (PET) and magnetic resonance imaging (MRI).[4–7] Several studies have explored these modalities to predict conversion from MCI to dementia.[8–12] Although these techniques provide useful information, they are invasive and expensive, limiting their applicability to well-resourced places and lacking the scalability and accessibility needed for low- and middle-income countries.[13] Furthermore, clinical and pathological variability is observed in AD using imaging techniques, which can make accurate diagnosis and prognosis challenging.[14]

In contrast, a neuropsychological test (NPT), conducted through an in-person interview, is currently the most accessible method for assessing cognitive decline. The NPT, triggered by patient history and in conjunction with a clinical examination, provides a comprehensive evaluation of cognitive function, including attention, memory, language, and visuospatial abilities. Researchers have explored computer-based approaches to predict the progression from MCI to AD using NPTs,[15–18] primarily relying on hand-crafted features and the cognitive scores extracted from the NPT by clinicians. However, these approaches have not yet achieved full automation, limiting their potential for more precise and efficient cognitive evaluations.

On the other hand, speech in the NPTs can be a strong predictor of cognitive decline,[19,20] and various artificial intelligence (AI)–assisted diagnostic models using linguistic and acoustic features extracted from the NPTs have been developed.[21–23] The Framingham Heart Study (FHS), which is the longest ongoing longitudinal, transgenerational cohort study of chronic disease, has been digitally recording the NPT interviews since 2005, and these voice recordings include all major established cognitive tests, such as the Boston Naming Test (BNT), Hooper Visual Organization Test, and Wechsler Memory Scale (WMS).[24] Several studies have used these recordings to develop diagnostic tools. For instance, a voice-based predictor was developed to identify dementia using acoustic features.[25] Xue et al. applied deep learning methods to acoustic features from FHS voice recordings to detect dementia and MCI.[26] In our earlier work, we used natural language processing (NLP) on the voice recordings to place each individual across the dementia spectrum.[27]

NLP, particularly large language models (LLMs) popularized with the introduction of ChatGPT, has emerged as a powerful tool in health care, showing reliable performance in various tasks.[28–30] By leveraging LLMs, we open up new frontiers in AD research, leading to the development of automated screening tools. Specifically, we consider the classification problem of determining whether individuals with MCI will progress to AD dementia within a 6-year window. Predicting conversions over a shorter period of time may be relatively easier, but has limited clinical utility.[31]

Our automated pipeline uses audio recordings of the NPT to predict the likelihood of MCI subjects transitioning to AD within 6 years. We emphasize that our analysis only uses text automatically transcribed from these recordings and it does not rely on any acoustic features. By leveraging transformer-based language models, we aim to capture semantic nuances potentially missed by conventional scoring, enriching the assessment with comprehensive text features. This underscores our plan for developing a cost-effective, automated tool that surpasses traditional methods in detecting AD progression. Conducting the NPT interview remotely, via a web interface without clinician participation, can further minimize screening costs. The pipeline incorporates diverse computational techniques, including speech recognition, speech diarization, a transformer-based sentence encoder, and logistic regression models.

RESEARCH IN CONTEXT

1. **Systematic review**: After conducting a systematic review of the literature, it is evident that no prior work has tried to automate the processing of voice recordings of neuropsychological tests using voice recognition to transcribe them into text and subsequently applying natural language processing (NLP) methods for analysis.
2. **Interpretation**: We have developed a novel approach to automate the prediction of progression to Alzheimer's disease (AD) within a 6-year timeframe using speech analysis. Our findings, derived from the neuropsychological test interviews conducted by the Framingham Heart Study, demonstrate strong performance, achieving an accuracy rate of 78.5% and a sensitivity of 81.1% in predicting progression to AD within 6 years.
3. **Future directions**: This study highlights the immense potential of integrating NLP techniques and speech data in predicting the future progression to AD. The method offers an opportunity to develop a cost-effective, widely accessible remote screening tool for predicting the progression to AD.

2 | METHODS

2.1 | Study participants

A cohort of 166 subjects with cognitive complaints were consecutively monitored by the FHS,[32] consisting of 59 males and 107 females, with a median age of 81 years (range: 63 to 97 years). It is noteworthy that

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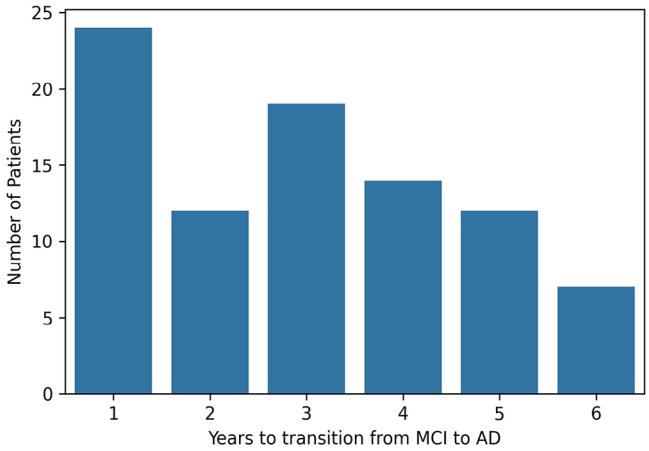

FIGURE 1 Number of MCI patients transitioning to AD annually over 6 years. AD, Alzheimer's disease; MCI, mild cognitive impairment.

the demographic composition of our cohort is predominantly White, reflecting the specific population from which the participants were drawn. Each participant underwent an approximately 1 hour long NPT, which was recorded and saved in the .wav format. The NPTs conducted by the FHS include subtests assessing different cognitive domains, such as memory, naming and language, visuoperceptual skills, abstract reasoning, and attention.[33,34] Additional information such as education, the type of apolipoprotein E (*APOE*) gene alleles, and health risk factors (such as blood glucose, diabetes, hypertension, etc.) were also available. All the participants have a completed NPT for which an MCI diagnosis was assigned. The cognitive status assignments such as AD diagnosis and MCI for those showing signs of cognitive decline was reached by consensus of at least one neurologist and one neuropsychologist, based on neurology exams, FHS study and external medical records, and brain imaging (the diagnostic procedure is outlined in Au et al.[33] and Satizabal et al.[35]). All participants have provided written informed consent and study protocols and consent forms were approved by the Boston University Medical Campus Institutional Review Board.

2.2 | Data preparation

The cohort for this study was derived from a larger group of participants whose NPTs were recorded by the FHS. This group consists of individuals at various cognitive stages, including some who have been diagnosed with MCI. Due to the increasing interest in AD and related clinical trials, our analysis focused on predicting the progression from MCI to AD. We elected not to consider progression from normal cognition to AD (or MCI) because the NPT has limited utility in predicting future cognitive decline in individuals without any current signs of cognitive deterioration. Therefore, we focused on MCI cases and identified those who had either progressed to AD or remained MCI within 6 years, as determined by a dementia review. Figure 1 shows the number of patients transitioning to AD from MCI each year over this period, representing the distribution of transitions, and indicating that a larger number of patients tend to transition to AD earlier within the 6-year timeframe. This observation suggests that the progression from MCI to AD is more likely to occur in the initial years after the MCI diagnosis.

In our previous work,[27] we developed a tool to automatically transcribe voice recordings. Each utterance was diarized (i.e., ascribed to a speaker: participant or examiner) and each transcript was split into the eight subtests comprising the FHS NPT. Some of these subtests are part of larger batteries of cognitive assessments such as WMS,[36] Wechsler Adult Intelligence Scale (WAIS),[37] and a revised form of the WAIS (WAIS-R).[38] In addition, there are several other tests that are frequently administered independently, including the BNT,[39] Verbal fluency (FAS),[40] and Clock Drawing Test (CDT).[41] The other two subtests are DEMO, which represents a part of the interview related to demographic information, and OTHER, which includes parts that are not categorized in the defined subtests. Using this developed tool, the participants' audio files were automatically transcribed, and each sentence was automatically labeled based on the specific subtest to which it belonged, such as WMS, WAIS, WAIS-R, BNT, FAS, CDT, DEMO, or OTHER. Figure 2 illustrates the automated pipeline to extract such structured data from the raw voice recording. From the prior study[27] leveraging a similar population, the diarization task demonstrated a performance with an exact F1 score of 70.2%, and the subtest classification task achieved an accuracy of 96.2%.

2.3 | Statistical analysis

The cohort consisted of 166 patients with MCI, 90 of whom progressed to AD dementia (progressive MCI) and 76 remained MCI (stable MCI) within the 6-year horizon. AD dementia included AD with stroke, AD without stroke, and mixed dementia (vascular + AD). Over a 6-year follow-up period, the participants with MCI had a mean (standard deviation) time to AD of 2.7 (1.5) years. Table 1 presents the participant characteristics, including self-reported sex, education status, age statistics, and six possible combinations of the three types of the *APOE* gene ($\varepsilon 2/\varepsilon 3/\varepsilon 4$) for both copies of the allele. The table suggests that older women with lower education levels and those carrying one or two copies of the *APOE* $\varepsilon 4$ allele are more likely to progress to AD. This finding aligns with previous studies that highlight age as the most significant risk factor for AD.[42] As individuals age, the prevalence of AD increases significantly, with estimates of 19% for those aged 75 to 84 and 30% to 35% for those > 85 years old.[43] Additionally, research shows that individuals who inherit one copy of the *APOE* $\varepsilon 4$ genotype have a higher risk of developing AD, while those who inherit two copies have an even higher risk.[44,45] Notably, in the progressive MCI group, females had an average age of 1.4 years older than males, suggesting that females may be more prone to progression due to their longer lifespan.

2.4 | Transcript encoding using universal sentence encoder

There are currently no standard methods for encoding a document into quantitative data. Based on selecting a specific segment of each

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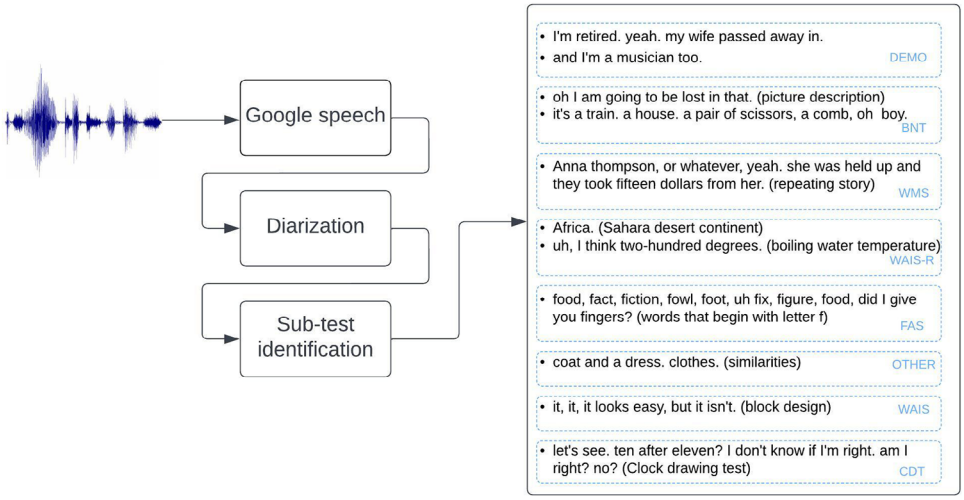

FIGURE 2 Automated pipeline for converting raw speech into structured data (as an example, the box on the right side contains a short note from each subtest highlighted in blue ink).

TABLE 1 Characteristics of patients with MCI, who either remain MCI or progress to AD within 6 years.

	Stable MCI *n* = 76	Progressive MCI *n* = 90	Difference
Age			
63–75	29	8	−21
75–85	36	44	not significant
85+	11	38	27
Sex (mean age)			
Female	44 [77.8]	63 [84.2]	19
Male	32 [77]	27 [82.8]	not significant
Education			
High school grad or less	33	46	13
Some college or more	43	44	not significant
APOE			
ε4/ε4	1	6	5
ε3/ε4 or ε2/ε4	19	29	10
ε2/ε2, ε3/ε3, or ε2/ε3	52	54	not significant

Abbreviations: AD, Alzheimer's disease; APOE, apolipoprotein E; MCI, mild cognitive impairment.

transcript, we obtain different vector embeddings for each NPT interview. To increase the training data, we randomly sample from each transcript to create several abbreviated versions that are then encoded. In addition, the content of each subtest can be encoded separately, resulting in eight specific embeddings. These embedding vectors are generated by a deep learning–based model, the Universal Sentence Encoder (USE).[46] The USE is a pretrained neural network based on the transformer architecture and has demonstrated a promising downstream classification accuracy on dementia detection and other tasks.[27,47] The USE outputs a 512-dimensional vector for each embedding. To simplify the downstream classification model, we perform dimensionality reduction using a logistic regression-based recursive feature elimination (RFE) method.[48] Specifically, we perform logistic regression-based RFE on the training data, systematically removing the weakest feature as determined by the smallest absolute value of the logistic regression coefficients.

2.5 | Prediction procedure

We generate deep learning–based embedding vectors from either an abbreviated version of a transcript or the content of one specific subtest. This results in eight embedding vectors associated with each subtest, as well as multiple embedding vectors from the abbreviated versions of one transcript. We then train a logistic regression model on the quantitative data associated with one subtest content, resulting in eight different trained models and eight scores for the subtests. However, the eight scores representing the subtests undergo a feature selection process using performance error analysis. The embeddings from multiple shortened versions of each transcript are treated as independent input, and one logistic regression model is trained on all of them, resulting in the generation of multiple scores for one transcript. Although the abbreviated versions of a transcript are treated independently during the embedding procedure, we take the average of the logistic regression scores to create the transcript average score (TAS). Finally, we feed the TAS score along with the selected subtest scores into an ensemble logistic regression model to make the final prediction of the likelihood of an individual with MCI converting to AD within 6 years. Figure 3 illustrates the prediction process. By integrating random abbreviation and subtest-specific embeddings through data augmentation, our approach significantly enhances the model's data interpretation and accuracy. This includes generating the TAS score from diverse transcript versions, alongside subtest evaluations to improve our prediction process. This strategy enriches our model's

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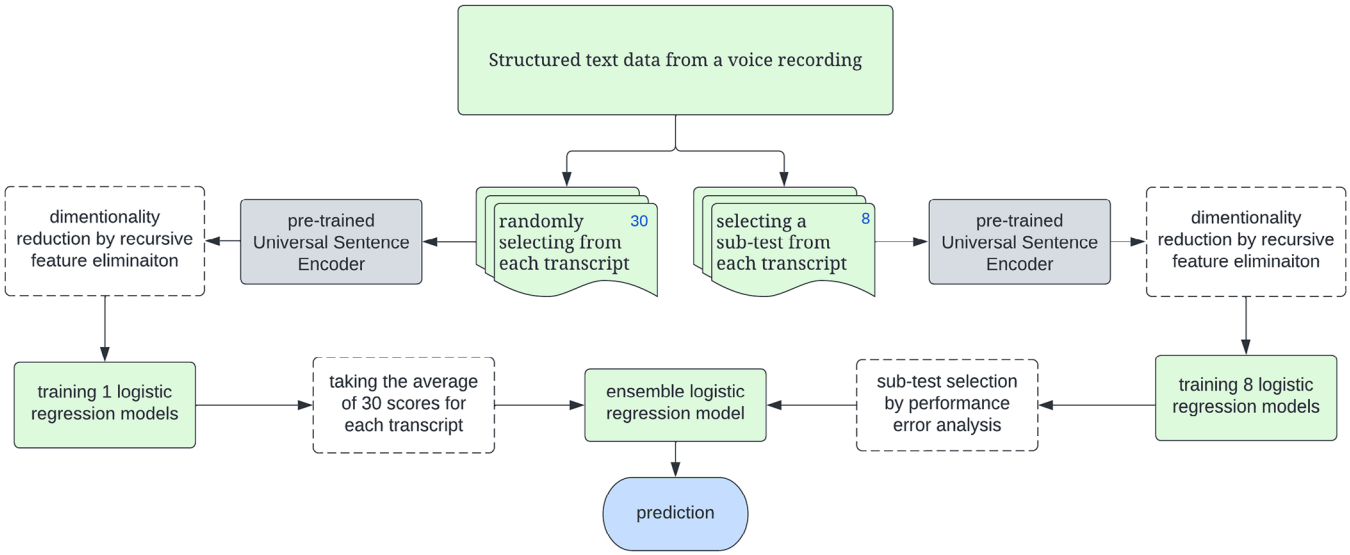

FIGURE 3 Automated pipeline for Alzheimer's disease prediction from a neuropsychological test interview.

data representation and predictive accuracy, leveraging both broad and detailed transcript insights.

2.6 | Validation and performance metrics

To evaluate our model's performance, we used a stratified group k-fold cross-validation approach, splitting the dataset into 10 folds. This division allocated 90% of the data for training (across nine folds) and 10% for testing (the remaining fold), with each segment serving as the test set once to ensure comprehensive evaluation. Within this framework, we also implemented an internal cross-validation within the training phase for dimensionality reduction and feature selection. This nested cross-validation strategy ensures the test data remain unseen until the final testing phase, enhancing the validity and reliability of our results. We conducted the stratified group k-fold cross-validation three times, each with a distinct random seed, to accurately calculate the average metrics and 95% confidence intervals for our model's performance assessment. The performance metrics considered for the evaluation were classification accuracy, sensitivity, specificity, precision, F1 score, and the area under the receiver operating characteristic curve (AUC). The AUC is a valuable measure that estimates the probability of the classifier ranking a randomly chosen progressive MCI subject (positive sample) higher than a randomly selected stable MCI subject (negative sample). Sensitivity and specificity provide insights into the correct classification of positive and negative subjects, while the F1 score measures the trade-off between precision and recall.

3 | RESULTS

Table 2 presents the average performance metrics of the logistic regression model, including the 95% confidence interval for each metric. The table is sorted in descending order based on AUC, with the highest value listed first. The first row showcases the model's performance, incorporating text, demographics, *APOE*, and health factors, achieving an AUC of 78.5% and an F1 score of 79.9%, marking the highest effectiveness observed. The subsequent two rows highlight models that leverage text features along with readily available demographic data such as age, sex, and education, also demonstrating strong predictive capabilities with an AUC and F1 score of 77.8% and 79.4% for our NLP model using only text features. The fourth row of the table reports the performance of adding *APOE* data to the model using demographic features, resulting in an AUC and F1 score of 71.7% and 75.7%. In addition, we trained a model with only demographic features as input, yielding an AUC of 68.8% as shown in row 6.

We also assessed a logistic regression model based on traditional neuropsychological test scores, including assessments like Logical Memory, Visual Reproductions, Paired Associate Learning Immediate Recall, Similarity Test, BNT, and Verbal Fluency Test. The model's performance, detailed in the fifth row, shows an AUC of 71.3% and an F1 score of 75.5%, underscoring that our NLP model not only matches but exceeds the predictive power of standard NPT scores. Additionally, when using four health factors (blood glucose, body mass index, presence of diabetes, and calculated low-density lipoprotein [LDL]) as input to the logistic regression, the seventh row shows an AUC of 66.2% and F1 score of 72.5%. As the Mini-Mental State Examination (MMSE) evaluates cognitive problems with thinking, communication, understanding, and memory, the model based on MMSE yielded an AUC of 60.7%. Other combinations of different features had no performance improvement over the best models in the first three rows of Table 2. Furthermore, the four health factors used in Table 2 (blood glucose, body mass index, presence of diabetes, and calculated LDL) resulted from the performance error analysis of 14 health factors; see the supporting information and Figure 4 for the complete analysis.

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TABLE 2 Average performance metrics (over 30 runs) on a held-out test set of the final logistic regression models using different features for MCI-to-AD progression in 6 years.

Features	AUC	Acc.	Sens.	Prec.	Spec.	F1 score
Text and demographics and *APOE* and health	78.5 (74.6, 82.5)	78.8 (75.6, 82.1)	80.6 (75.9, 85.2)	80.4 (76.8, 84.1)	76.9 (72.2, 81.5)	79.9 (74.6, 82.5)
Text	77.8 (74.2, 81.3)	78.2 (75.0, 81.4)	81.1 (75.9, 86.3)	78.9 (75.8, 81.9)	75.0 (70.9, 79.1)	79.4 (76.0, 82.7)
Text and demographics	77.5 (73.8, 81.2)	78.5 (75.4, 81.7)	81.1 (75.8, 86.3)	79.3 (76.1, 82.6)	75.6 (71.4, 79.8)	79.6 (76.1, 83.0)
Demographics and *APOE*	71.7 (67.7, 75.6)	74.4 (71.9, 76.9)	77.8 (71.5, 84.1)	77.1 (73.3, 80.9)	70.6 (63.6, 77.6)	75.7 (72.7, 78.7)
Traditional NP tests	71.3 (67.2, 75.5)	74.7 (71.8, 77.6)	77.2 (70.7, 83.7)	77.2 (73.5, 80.8)	71.9 (66.3, 77.5)	75.5 (72.0, 79.0)
Demographics	68.8 (64.3, 73.3)	70.6 (67.1, 74.1)	70.6 (64.5, 76.6)	74.9 (70.4, 79.4)	70.6 (64.3, 77.0)	71.1 (67.5, 74.8)
Health factors	66.2 (63.1 71.2)	71.2 (68.2, 74.1)	75.0 (68.4, 81.6)	73.2 (69.7, 76.7)	66.9 (61.2, 72.5)	72.5 (68.9, 76.1)
MMSE	60.7 (55.9, 65.4)	62.9 (59.5, 64.4)	66.7 (60.8, 72.6)	65.2 (61.4, 69.0)	58.8 (52.9, 64.6)	64.9 (61.1, 68.8)

Abbreviations: Acc., accuracy; *APOE*, apolipoprotein E; AUC, area under the receiver operating characteristic curve; MMSE, Mini-Mental State Examination; NP, neuropsychological; Prec., precision; Sens., sensitivity; Spec., specificity.

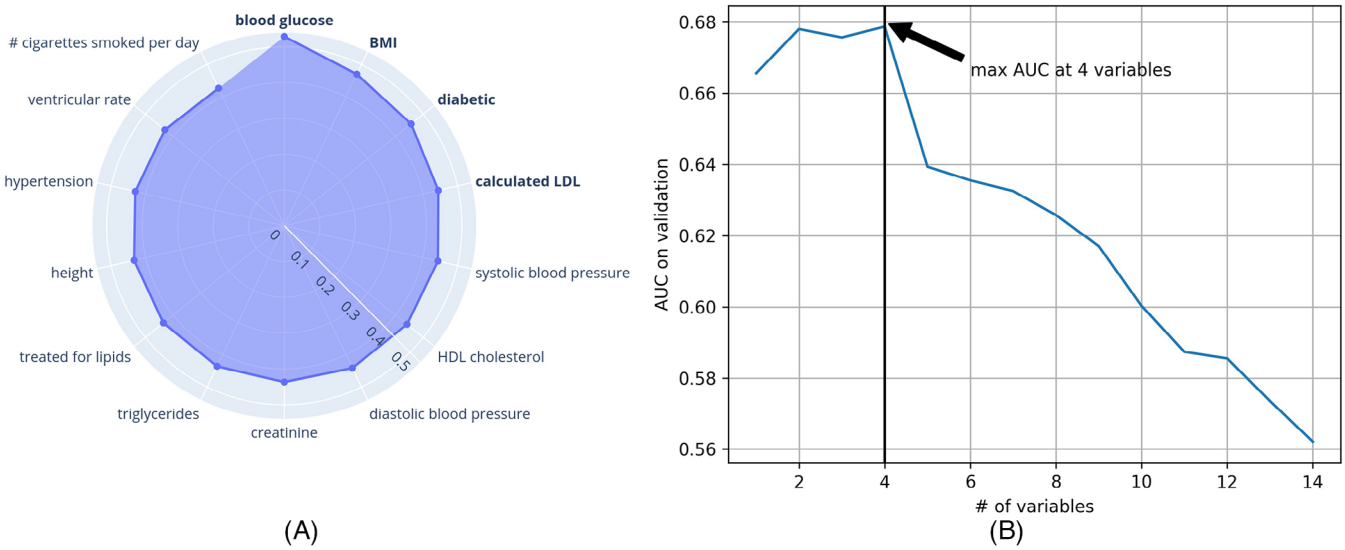

FIGURE 4 Performance error analysis for health factors. A, Performance error (1-AUC) after removing each feature at a time. B, Results of AUC for an arbitrary number of most important features. AUC, area under the receiver operating characteristic curve; BMI, body mass index; LDL, low-density lipoprotein.

Based on the confidence intervals detailed in Table 2, the performance metrics of the first three rows, which use the text feature set, distinguish them significantly from other models presented in the table. While there may be some overlap in confidence intervals between models using text features and baseline models, statistical analysis, such as the paired *t* test, validates that the AUC for models using text features is significantly improved, underscoring the efficacy of our NLP approach in enhancing predictive accuracy.

Figure 5 displays the coefficients of our logistic regression model using the text features and the demographics model output. The results have been adjusted for continuous variables through *z* score normalization (by subtracting the mean and dividing by the standard deviation), making the coefficients comparable. This figure represents the distribution of logistic regression coefficients for different features, highlighting their relative importance in the model's predictive process. By comparing the interquartile ranges and medians of coefficients for TAS and selected subtests against the demographic features, we can observe a difference in their contributions. A higher median value for TAS and subtests implies these variables have a stronger predictive value, underscoring their role over demographic factors in influencing the model's prediction.

4 | DISCUSSION

Speech during cognitive exams has been identified as a promising biomarker that strongly correlates with underlying cognitive dysfunction. The current study aimed to automatically predict the progression to AD using NLP and machine learning techniques applied to speech data. The proposed method predicted the participant's progression to

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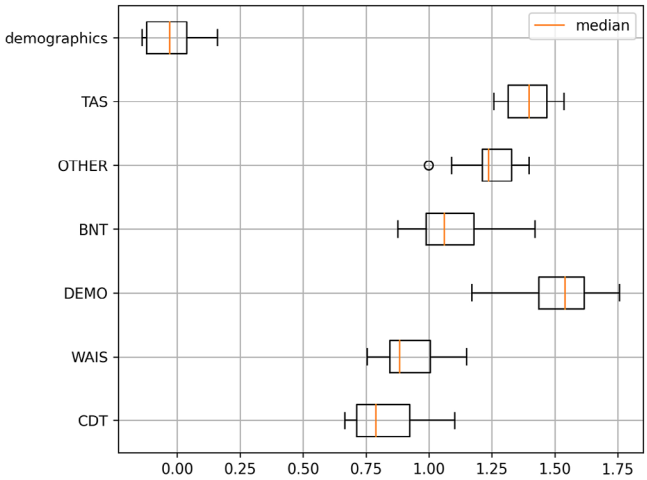

FIGURE 5 Logistic regression coefficients of the text features and demographics used in the proposed method. Demographics includes age, sex, and education. BNT, Boston Naming Test; CDT, Clock Drawing Test; DEMO, part of the interview related to demographic information; OTHER, similarity tests; TAS, transcript average score; WAIS, Wechsler Adult Intelligence Scale.

AD with an accuracy of 78.2% and a sensitivity of 81.1% in the held-out test data, demonstrating strong predictive power over a 6-year span. However, the specificity of predicting whether an individual with MCI will progress to AD within 6 years was moderate, at 75%. To reduce the costs associated with recruiting subjects for clinical trials, it is important to improve the specificity. Nevertheless, the relatively high sensitivity of our prediction tool makes it clinically applicable and potentially beneficial for eventual neuroprotective therapies.[49]

Importantly, our method only uses features derived from speech data in an automated manner, along with easily obtainable variables such as age, sex, and education level. The proposed method offers a non-invasive, accessible, and easy-to-administer AI-based predictive approach because it does not require data involving laboratory tests, genetic tests, or imaging exams. This makes it a promising candidate for integration into remote assessment technologies. A major strength of this study is its use of semantic features extracted from the structured text data. This approach allows for the potential transferability of the entire pipeline to other languages, leveraging the availability of transcription tools that can transcribe from any language to English, and/or powerful NLP models in different languages.[50,51] As a computer-aided decision-making tool, our method has the potential to mitigate inter-clinician variability in selecting candidates for clinical trials and drug tests, enhancing the consistency and reliability of participant selection processes.[52]

The Results section indicates that adding demographic features to text features does not enhance the model's ability to predict the progression from MCI to AD. This contrasts with previous assumptions about the predictive power of age and other demographics in relation to degenerative diseases over extended periods. Even though there are significant differences in demographics between stable and progressive MCI groups, the use of text features alone outperforms the use of demographic features. This underscores the strong predictive strength of the engineered text features. Moreover, upon evaluating the performance of the logistic regression model using the traditional NPT scores, we observed an AUC of 71.3%. This result indicates that our approach outperforms conventional NPT scoring methods in this study. Furthermore, when we compared our model to a well-established cognitive assessment score such as the MMSE score, text features demonstrated higher predictive power. In addition, compared to other works that used only non-invasive features,[53,54] our model's F1 score = 79.4% is higher. For instance, the authors in one paper[53] predicted AD transition within 9 years based on NPT scores provided by specialized clinicians, achieving an F1 score of 70.8%, whereas Grassi et al. achieved an F1 score of 72.7% using sociodemographic characteristics, clinical information, and NPT scores.[54] These methods still require highly specialized personnel to generate the NPT scores while our method is fully automated, making AD prediction accessible to all.

As depicted in Figure 5, our analysis revealed that subtests related to demographic questions (DEMO), BNT, similarity tests (OTHER), and WAIS emerged as the top features driving the performance of our model. These sections of each transcript are key predictors for identifying the future incidence of AD. Thus, our approach facilitates the identification of subtests that provide more informative insights for predicting the future incidence of AD. This finding underscores the potential benefit of using a more structured interview to better capture the language deficits that may underlie cognitive decline. Additionally, after conducting a performance error analysis on 14 health risk factors, we found that variables such as blood glucose, body mass index, diabetes, and calculated LDL were useful in predicting the development of AD. In conclusion, our study demonstrates the potential of using automatic speech recognition and NLP techniques to develop a prediction tool for identifying individuals with MCI who are at risk of developing AD. Our method achieved high accuracy and outperformed other non-invasive approaches. However, further prospective studies with larger populations are necessary to validate the generalizability of our models. Additionally, it is important to standardize the definition of MCI across different locations to enable better comparison of results. With continued development and refinement, our approach may contribute to early intervention and selection in clinical trials for novel AD treatments, ultimately improving patient outcomes.

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CONFLICT OF INTEREST STATEMENT

Rhoda Au is a scientific advisor to Signant Health and NovoNordisk and consultant to Biogen and the Davos Alzheimer's Collaborative. She receives funding from the National Institute on Aging (AG072654,

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AG062109, AG068753) and has also been supported through awards from the American Heart Association, the Alzheimer's Drug Discovery Foundation, Alzheimer's Disease Data Initiative, and Gates Ventures. Vijaya B. Kolachalama has received support from the Karen Toffler Charitable Trust; Johnson & Johnson (through the Boston University Lung Cancer Alliance); the NIH under grants RF1-AG062109, R01-HL159620, R43-DK134273, and R21-CA253498; the American Heart Association under grant 20SFRN35460031; and serves as a consultant to AstraZeneca. Both R. Au and V. B. Kolachalama state no conflicts of interest with the present work. There is no declaration from other authors. Author disclosures are available in the supporting information.

CONSENT STATEMENT

All participants have provided written informed consent and study protocols and consent forms were approved by the Boston University Medical Campus Institutional Review Board.

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REFERENCES

1. Mitchell AJ, Shiri-Feshki M. Rate of progression of mild cognitive impairment to dementia–meta-analysis of 41 robust inception cohort studies. *Acta Psychiatr Scand*. 2009;119:252-265.
2. Liu S, Cao Y, Liu J, Ding X, Coyle D, Initiative ADN. A novelty detection approach to effectively predict conversion from mild cognitive impairment to Alzheimer's disease. *Int J Mach Learn Cybern*. 2023;14:213-228.
3. Pereira T, Ferreira FL, Cardoso S, et al. Neuropsychological predictors of conversion from mild cognitive impairment to Alzheimer's disease: a feature selection ensemble combining stability and predictability. *BMC Med Inform Decis Mak*. 2018;18:1-20.
4. Scheltens P, Blennow K, Breteler M, et al. Alzheimer's disease. *Lancet (Lond Engl)*. 2016;388:505-517.
5. Turner RS, Stubbs T, Davies DA, Albensi BC. Potential new approaches for diagnosis of alzheimer's disease and related dementias. *Front Neurol*. 2020;11:496.
6. Thomas JA, Burkhardt HA, Chaudhry S, et al. Assessing the utility of language and voice biomarkers to predict cognitive impairment in the Framingham Heart Study cognitive aging cohort data. *J Alzheimers Dis*. 2020;76:905-922.
7. Weiner MW, Veitch DP, Miller MJ, et al. Increasing participant diversity in AD research: plans for digital screening, blood testing, and a community-engaged approach in the Alzheimer's Disease Neuroimaging Initiative 4. *Alzheimers Dement*. 2023;19:307-317.
8. Caminiti SP, Ballarini T, Sala A, et al. FDG-PET and CSF biomarker accuracy in prediction of conversion to different dementias in a large multicentre MCI cohort. *NeuroImage Clin*. 2018;18:167-177.
9. Long X, Chen L, Jiang C, Zhang L, Initiative ADN. Prediction and classification of Alzheimer disease based on quantification of MRI deformation. *PloS One*. 2017;12:e0173372.
10. Varatharajah Y, Ramanan VK, Iyer R, Vemuri P. Predicting short-term MCI-to-AD progression using imaging, CSF, genetic factors, cognitive resilience, and demographics. *Sci Rep*. 2019;9:2235.
11. Ahmadzadeh M, Christie GJ, Cosco TD, Moreno S. Neuroimaging and analytical methods for studying the pathways from mild cognitive impairment to Alzheimer's disease: protocol for a rapid systematic review. *Syst Rev*. 2020;9:1-6.
12. Ritter K, Schumacher J, Weygandt M, et al. Multimodal prediction of conversion to Alzheimer's disease based on incomplete biomarkers. *Alzheimers Dement Diagn Assess Dis Monit*. 2015;1:206-215.
13. Clute-Reinig N, Jayadev S, Rhoads K, Le Ny A-L. Alzheimer's disease diagnostics must be globally accessible. *J Alzheimers Dis*. 2021;84:1453-1455.
14. Kelley S, Perez-Urrutia N, Morales R. Misfolded amyloid-β strains and their potential roles in the clinical and pathological variability of Alzheimer's disease. *Neural Regen Res*. 2023;18:119.
15. Tabert MH, Manly JJ, Liu X, et al. Neuropsychological prediction of conversion to Alzheimer disease in patients with mild cognitive impairment. *Arch Gen Psychiatry*. 2006;63:916-924.
16. Chapman RM, Mapstone M, McCrary JW, et al. Predicting conversion from mild cognitive impairment to Alzheimer's disease using neuropsychological tests and multivariate methods. *J Clin Exp Neuropsychol*. 2011;33:187-199.
17. Silva D, Guerreiro M, Santana I, et al. Prediction of long-term (5 years) conversion to dementia using neuropsychological tests in a memory clinic setting. *J Alzheimers Dis*. 2013;34:681-689.
18. Pereira T, Lemos L, Cardoso S, et al. Predicting progression of mild cognitive impairment to dementia using neuropsychological data: a supervised learning approach using time windows. *BMC Med Inform Decis Mak*. 2017;17:1-15.
19. Stück D, Signorini A, Alhanai T, et al. Novel digital voice biomarkers of dementia from the Framingham Study. *Alzheimers Dement*. 2018;14:P778-P779.
20. Boschi V, Catricala E, Consonni M, Chesi C, Moro A, Cappa SF. Connected speech in neurodegenerative language disorders: a review. *Front Psychol*. 2017;8:269.
21. Hernández-Domínguez L, Ratté S, Sierra-Martínez G, Roche-Bergua A. Computer-based evaluation of Alzheimer's disease and mild cognitive impairment patients during a picture description task. *Alzheimers Dement Diagn Assess Dis Monit*. 2018;10:260-268.
22. Liu L, Zhao S, Chen H, Wang A. A new machine learning method for identifying Alzheimer's disease. *Simul Model Pract Theory*. 2020;99:102023.
23. Pulido MLB, Hernández JBA, MÁF Ballester, González CMT, Mekyska J, Smékal Z. Alzheimer's disease and automatic speech analysis: a review. *Expert Syst Appl*. 2020;150:113213.
24. Downer B, Fardo DW, Schmitt FA. A summary score for the Framingham Heart Study neuropsychological battery. *J Aging Health*. 2015;27:1199-1222.
25. Lin H, Karjadi C, Ang TF, et al. Identification of digital voice biomarkers for cognitive health. *Explor Med*. 2020;1:406.
26. Xue C, Karjadi C, Paschalidis IC, Au R, Kolachalama VB, Detection of dementia on raw voice recordings using deep learning: A Framingham Heart Study. Available SSRN 3788945 2021.
27. Amini S, Hao B, Zhang L, et al. Automated detection of mild cognitive impairment and dementia from voice recordings: a natural language processing approach. *Alzheimers Dement*. 2022.
28. Iroju OG, Olaleke JO. A systematic review of natural language processing in healthcare. *Int J Inf Technol Comput Sci*. 2015;7:44-50.
29. Srivastava SK, Singh SK, Suri JS. A healthcare text classification system and its performance evaluation: a source of better intelligence by characterizing healthcare text. *Cogn. Inform. Comput. Model. Cogn. Sci.*. Elsevier; 2020:319-369.
30. Robin J, Xu M, Balagopalan A, et al. Characterizing progressive speech changes in prodromal-to-mild Alzheimer's disease using natural language processing. *Alzheimers Dement*. 2022;18:e063244.
31. Chen J, Chen G, Shu H, et al. Predicting progression from mild cognitive impairment to Alzheimer's disease on an individual subject basis by applying the CARE index across different independent cohorts. *Aging*. 2019;11:2185.

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32. Andersson C, Johnson AD, Benjamin EJ, Levy D, Vasan RS. 70-year legacy of the Framingham Heart Study. *Nat Rev Cardiol*. 2019;16:687-698.
33. Au R, Piers RJ, Devine S. How technology is reshaping cognitive assessment: lessons from the Framingham Heart Study. *Neuropsychology*. 2017;31:846.
34. Jak AJ, Preis SR, Beiser AS, et al. Neuropsychological criteria for mild cognitive impairment and dementia risk in the Framingham Heart Study. *J Int Neuropsychol Soc JINS*. 2016;22:937.
35. Satizabal CL, Beiser AS, Chouraki V, Chêne G, Dufouil C, Seshadri S. Incidence of dementia over three decades in the Framingham Heart Study. *N Engl J Med*. 2016;374:523-532.
36. Wechsler D, Scale-Revised WAI, The psychological corporation. San Antonio TX. 1997.
37. Wechsler D, The measurement and appraisal of adult intelligence, 1958. Baltim Wiliams Wilkins 2020.
38. Zarantonello MM, Munley PH, Milanovich J. Predicting wechsler adult intelligence scale-revised (WAIS-R) IQ scores from the luria-nebraska neuropsychological battery (form I). *J Clin Psychol*. 1993;49:225-233.
39. Goodglass H, Kaplan E. *The assessment of aphasia and related disorders*. 1972.
40. Franzen MD. Multilingual aphasia examination. *Kans City MO Test Corp Am*. 1986.
41. Amini S, Zhang L, Hao B, et al. An artificial intelligence-assisted method for dementia detection using images from the clock drawing test. *J Alzheimers Dis*. 2021;83:581-589.
42. Herrup K. Reimagining Alzheimer's disease—an age-based hypothesis. *J Neurosci*. 2010;30:16755-16762.
43. Armstrong RA. Risk factors for Alzheimer's disease. *Folia Neuropathol*. 2019;57:87-105.
44. Farrer LA, Cupples LA, Haines JL, et al. Effects of age, sex, and ethnicity on the association between apolipoprotein E genotype and Alzheimer disease: a meta-analysis. *Jama*. 1997;278:1349-1356.
45. Liu C-C, Kanekiyo T, Xu H, Bu G. Apolipoprotein E and Alzheimer disease: risk, mechanisms and therapy. *Nat Rev Neurol*. 2013;9:106-118.
46. Cer D, Yang Y, Kong S, et al. Universal sentence encoder. ArXiv Prepr ArXiv180311175. 2018.
47. Vaswani A, Shazeer N, Parmar N, et al. Attention is all you need. ArXiv Prepr ArXiv170603762. 2017.
48. Hao B, Sotudian S, Wang T, et al. Early prediction of level-of-care requirements in patients with COVID-19. *Elife*. 2020;9:e60519.
49. Eskildsen SF, Coupé P, García-Lorenzo D, et al. Prediction of Alzheimer's disease in subjects with mild cognitive impairment from the ADNI cohort using patterns of cortical thinning. *Neuroimage*. 2013;65:511-521.
50. Yang Y, Cer D, Ahmad A, et al. Multilingual universal sentence encoder for semantic retrieval. ArXiv Prepr ArXiv190704307. 2019.
51. Conneau A, Khandelwal K, Goyal N, et al. Unsupervised cross-lingual representation learning at scale. ArXiv Prepr ArXiv191102116. 2019.
52. Martínez-Nicolás I, Llorente TE, Martínez-Sánchez F, Meilán JJG. Ten years of research on automatic voice and speech analysis of people with Alzheimer's disease and mild cognitive impairment: a systematic review article. *Front Psychol*. 2021;12:620251.
53. Merone M, D'Addario SL, Mirino P, et al. A multi-expert ensemble system for predicting Alzheimer transition using clinical features. *Brain Inform*. 2022;9:20.
54. Grassi M, Rouleaux N, Caldirola D, et al. A novel ensemble-based machine learning algorithm to predict the conversion from mild cognitive impairment to Alzheimer's disease using socio-demographic characteristics, clinical information, and neuropsychological measures. *Front Neurol*. 2019;10:756.

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